

Choose one route | Independent practice

Name: _____ Class: _____ Date: _____

Answer the original question extracts on the following pages. Keep the original question numbers and include the route letter. Choose one route with your teacher, not all three.

Route	Topics	Marks	Booklet pages
A	Embedded systems and networks	10	2-4
B	Cyber security and databases	11	5-7
C	Core programming and validation	12	8-10

Aim for about 13-14 minutes of independent work, then check your first attempt with the marking sheet your teacher provides. If you need more time, record unfinished parts as not attempted. Do not replace a first answer with a corrected answer before recording the original score.

You may work on this paper or use the digital notebook. For a flowchart, use the supplied drawing space and upload a photograph. For a trace, use the printed table or type values into the digital table.

The routes assess different skills and have different totals; do not compare them as grades.

Reproduced from the OxfordAQA g210/2 papers supplied for classroom use. Original question wording, diagrams, tables and mark allocations are retained.

Route A | Embedded systems + networks

Source: 2019-06 Paper 2, original page 21 | original question numbers retained

1 3 . 1 What is an embedded system?

[1 mark]

1 3 . 2 Describe **two** differences between an embedded system and a standard desktop computer.

[2 marks]

Difference 1

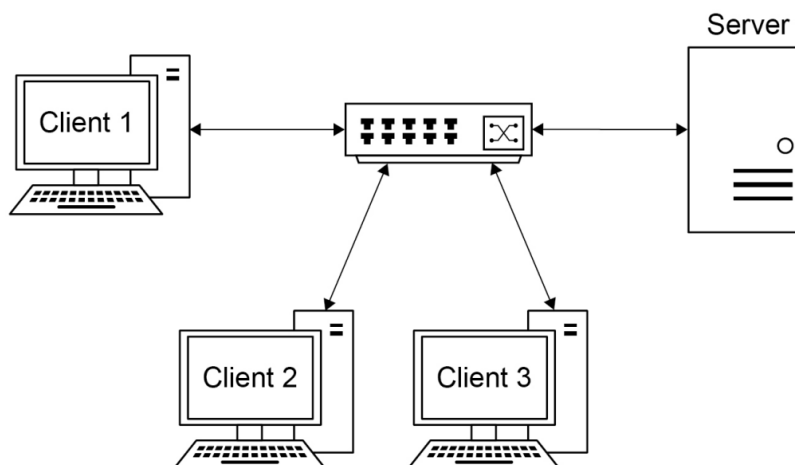
Difference 2

Route A | Embedded systems + networks

Source: 2019-06 Paper 2, original page 4 | original question numbers retained

0 3 . 1 A diagram of part of a school's computer network is shown in **Figure 2**.

Figure 2



Identify the physical network topology shown in **Figure 2**.

[1 mark]

0 3 . 2 WANs and LANs are two different types of network.

State **one** difference between a WAN and a LAN.

[1 mark]

Route A | Embedded systems + networks

Source: 2022-06 Paper 2, original page 18 | original question numbers retained

1	0
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1

 Explain what a computer network is.

[1 mark]

1 0 . 2 Describe the benefits and risks of computer networks.

[4 marks]

[illegible]

Route B | Cyber security + databases

Source: 2020-11 Paper 2, original page 14 | original question numbers retained

10

A relational database stores information about the products that a shop sells and the sales that have been made.

The database contains two tables: **Product** and **Sale**.

Some of the contents of these tables are shown in **Figure 2**. Note that not all of the records in the tables are shown.

Figure 2

Product

ProductID	Description	Category	PriceEach	NumberInStock
108	Pencil	Stationery	1	0
203	Ruler	Stationery	2	10
297	Orange Juice	Drinks	3	23
406	Water	Drinks	2	15
556	Bread	Food	2	10

Sale

SaleID	Date	ProductID	QuantitySold
S1201	05/08/19	203	1
S1202	05/08/19	406	2
S1203	05/08/19	108	1

10.1

Explain what a foreign key is and state the name of the field that has been used as a foreign key in this database.

[2 marks]

Route B | Cyber security + databases

Source: 2020-11 Paper 2, original page 15 | original question numbers retained

102

Write an SQL query to search the database and list all the sales of drinks.
For each sale the SaleID, ProductID, Description and QuantitySold should be listed.

[4 marks]

SELECT

FROM

WHERE

103

When the shop receives new stock the values in the NumberInStock field of the **Product** table must be updated.

Tick (✓) **one** box to show which SQL query is the correct one to change the NumberInStock of ProductID 203 from 10 to 15

[1 mark]

SQL Query	Tick one
UPDATE NumberInStock = 15 FROM Product WHERE ProductID = 203	
UPDATE NumberInStock + 5 FROM Product WHERE ProductID = 203	
UPDATE Product SET NumberInStock = 15 WHERE NumberInStock = 10	
UPDATE Product SET NumberInStock = 15 WHERE ProductID = 203	

Route B | Cyber security + databases

Source: 2024-06 Paper 2, original page 21 | original question numbers retained

10

A computer virus is one type of malware.

10.1

Describe what a computer virus is.

[3 marks]

10.2

State **one** other type of malware.

[1 mark]

Route C | Core programming + validation

Source: 2022-06 Paper 2, original page 11 | original question numbers retained

07

Figure 3 shows an algorithm expressed using pseudocode.

The algorithm operates on an array called `List`. The current contents of the array `List` are shown in Figure 4.

Figure 3

```
Result ← 0
Num ← USERINPUT
FOR Count ← 0 TO LEN(List) - 1
    IF Num > List[Count] THEN
        Temp ← Num - List[Count]
    ELSE
        Temp ← List[Count] - Num
    ENDIF
    IF Temp > Result THEN
        Result ← Temp
    ENDIF
ENDFOR
```

Figure 4

Index	[0]	[1]	[2]
Contents	4	9	6

07.1

Complete the trace table below to show the execution of the algorithm in Figure 3 on the array contents in Figure 4. Assume that the user enters the number 5

You may not need to write in all of the rows of the table.

[5 marks]

Result	Count	Temp

07.2

What is the purpose of the algorithm shown in Figure 3?

[1 mark]

Route C | Core programming + validation

Source: 2023-11 Paper 2, original page 2 | original question numbers retained

Answer **all** questions in the spaces provided.

0 1 . 1 **Table 1** lists the definitions of three terms.

Complete **Table 1** by stating the names of the terms that are defined.

[2 marks]

Table 1

Definition of term	Term name
The process of removing unnecessary details from a problem.	
A sequence of steps that can be followed to complete a task.	
Breaking a problem down into sub problems so that each sub problem accomplishes an identifiable task.	

0 1 . 2 A computer program needs to accept the input of the percentage mark that a student has achieved in a test. The percentage mark must be between 0 and 100 inclusive.

In the box on the **next page**, draw a flowchart for an algorithm that:

- displays a message asking the user to input a mark
- gets the user to input a mark
- if the mark is less than 0 or above 100, then:
 - the error message `Invalid mark` should be displayed
 - the user should be made to input a mark again until the mark is in the allowed range
- displays the message `Mark accepted` and then ends once a mark in the allowed range has been entered.

[4 marks]

Do not write in this space

Route C | Core programming + validation

Source: 2023-11 Paper 2, original page 3 | original question numbers retained

DRAW FLOWCHART FOR QUESTION 01.2 HERE

Turn over for the next question